

DATA PROFESSIONALS SURVEY DASHBOARD

Project Title: Professional Survey Analysis

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Background Study

This project analyzes a survey dataset of data professionals to understand:

- Salary distribution
- Job roles
- Programming language preferences
- Work satisfaction levels
- Geographic distribution

The goal is to derive insights into career trends within the data industry.

Problem Statement

This project aims to analyze survey data from data professionals to uncover trends in salary distribution, programming language usage, job roles and job satisfaction across different regions and experience levels.

Objectives

- Organizations and aspiring data professionals need insight into:
- Which roles earn the highest salaries
- Most commonly used programming languages
- Job satisfaction across roles
- Work/life balance ratings
- Regional salary differences

Data Description

The dataset contains survey responses including:

- Job Title
- Country

- Annual Salary
- Programming Language
- Work Satisfaction
- Work-Life Balance
- Industry
- Years of Experience

Methodology

Data Cleaning Process (Power Query)

The following transformations were performed:

- Removed some blank rows
- Standardized job titles
- Converted salary column to numeric format
- Split multi-select programming language column
- Removed duplicates
- Renamed columns for clarity
- Replaced null values where necessary

Analysis

Analysis Questions

- Develop insights into the loan default rates by customer demographics (e.g., Income, employment length). The rate at which borrowers default should be looked at based on their different demographics like their annual income.
- Analyze the interest rate and their impact on loan performance.
- The loan amount collected by different customer segments and the loan purposes. The amount of loans collected by a particular segment and the reason why they are collecting the loan should be analyzed.

- The loan approval rates by region and customer risk profiles should be analyzed accordingly.
- Develop insights into high-risk loans by analyzing the relationship credit scores, loan amount and interest rate.
- Identify patterns in loan approval rates based on borrower demographics and loan conditions.

Analysis Approach

A star schema approach was used:

- Fact Table: Survey Responses
- Dimension Fields: Country, Job Title, Programming Language

Basic DAX measures created:

```
“
Total Respondents = COUNT(Survey[Respondent ID])

Average Salary = AVERAGE(Survey[Salary])

Average Satisfaction = AVERAGE(Survey[Job Satisfaction])
“
```

Dashboard Overview

- Total Respondents
- Average Salary
- Most Common Job Role
- Most Used Programming Language
- Salary by Job Title
- Salary by Country
- Salary by Experience Level
- Programming Language Popularity
- Language Preference by Role
- Work-Life Balance Ratings
- Salary vs Satisfaction (Scatter Plot)
- Satisfaction by Role

Analysis Findings

- Python is the dominant programming language.

- Data Scientists and Engineers earn higher average salaries.
- Experience level positively impacts salary.
- Satisfaction varies more by role than by salary.
- Some regions show higher compensation concentration.
- Higher salaries do not always correlate with higher job satisfaction.

Recommendation

- Companies should align compensation with market trends.
- Investing in Python-based skill development increases employability.
- Monitoring employee satisfaction can reduce attrition risk.
- Regional salary benchmarking improves competitive positioning.



